

Project Report
RISETECH COMPUTER ENGINEERING
DEPARTMENT NUST EME COLLEGE(ORANIZATION)
Health Safety And Environment



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ABSTRACT

In this semester project our organization **RISETECH COMPUTER ENGINEERING DEPARTMENT NUST EME COLLEGE**. Under the dispensation of the HSE development program of the college of electrical and mechanical engineering-national university of sciences and technology, this group audit for RISETECH is cross-sectional and aimed at auditing the health, safety, and environmental practices within the organization. Compliance to international best practice by way of the iso 45001:2018 occupational health and safety management systems (OHSMS) and iso 14001:2015 environmental management systems (EMS) standards is strictly followed during the audit, which also touches compliance with other key area audits that will include identifying hazards in the workplace, assessing risks, and legal and regulatory compliance, along with a review of existing safety and environmental controls. Several non-conformities and areas for improvement are cited as a result of comprehensive site surveys, document reviews, and stakeholder interviews. Proposed actions include those available for mitigation and control measures to eliminate or reduce occupational hazards, environ-sensitive impacts, and therefore internalizing appropriate culture within an organization that thrives on proactive HSE-modelled operations. This audit strengthens RISETECH'S commitment to sustainability, while, at the same time, serving as a good experience for engineering students to apply iso standards in real-life situations.

CHAPTER ONE

1. INTRODUCTION OF THE ORGANIZATION

1.1 INTRODUCTION:

RISETECH, a research unit within the computer engineering department at NUST college of EME, concentrates on academic research, prototyping embedded systems, developing IoT solutions, and integrating hardware and software. It offers a diverse learning environment that combines electronics, programming, and systems design, providing areas for hardware prototyping, software development, and embedded system research.

1.2 PURPOSE OF THE AUDIT

The health, safety, and environment (HSE) audit evaluates RISETECH workplace safety practices and environmental compliance, identifying potential hazards, assessing risks, and recommending mitigation strategies in accordance with ISO 45001:2018 and ISO 14001:2015 standards, ensuring personnel safety and sustainable environmental practices.

1.3 OBJECTIVES OF THE AUDIT

The main goals of this report are to:

- Examine the present health, safety, and environmental conditions within the research and development facility.
- Recognize and assess potential risks to health, safety, and the environment that may arise from regular operations, equipment usage, and workspace conditions.
- Assess the lab's existing procedures in light of the requirements outlined in ISO 45001 and ISO 14001, identifying both its strengths and areas that require enhancement.
- Suggest practical, affordable measures to mitigate or eliminate hazards, ensuring enhanced safety, reduced environmental impact, and long-term sustainability.

1.4 SCOPE OF THE AUDIT

The audit covers all aspects of RISETECH'S operations and functions, with a particular emphasis on:

- Workplaces and administrative areas.
- Server rooms and IT infrastructure zones.
- Research facilities.
- Workshops and equipment storage locations.

These regions are evaluated for potential risks, compliance gaps, emergency readiness, and overall adherence to HSE best practices.

CHAPTER TWO

2. GENERAL OPERATIONAL COMPONENTS

2.1 GENRAL OPERATIONAL COMPONENTS

The research center includes multiple separate areas:

- **Labs and Hardware Prototyping Stations:**
This tool is employed for circuit design, sensor integration, and practical testing of embedded devices.
- **Software Development and Testing Zones:**
The system is furnished with desktop systems for coding, simulation, and debugging software components.
- **Soldering and PCB Manufacturing Area:**
The system is furnished with desktop systems for coding, simulation, and debugging software components.
- **Power Supply Units and Test Benches:**
The product comprises programmable power supplies, multimeters, oscilloscopes, and electronic loads.
- **Storage for Electronic Components and Chemicals:**
Includes cabinets for microcontrollers, sensors, resistors, and chemical agents utilized in the cleaning of pcb components.
- **Server and Networking Racks:**
It encompasses the provision of backend services, hosting in local areas, and the management of iot cloud platforms.
- **Waste Disposal Area (E-waste and General Waste):**
The person is accountable for disposing of different lab waste, such as printed circuit boards, resistors, batteries, and general lab waste.

CHAPTER THREE

3. WORKPLACE-WISE HAZARDS IDENTIFICATION AND RISK ASSESSMENT

3.1 WORKPLACE-WISE HAZARDS IDENTIFICATION AND RISK ASSESSMENT

Table 3.1: WORKPLACE-WISE HAZARDS IDENTIFICATION AND RISK ASSESSMENT

Workplace	Hazards Identified	Causes	Effects
Soldering Station	Fume inhalation, burns, fire hazard	Poor ventilation, mishandling of hot tools	Respiratory problems, skin injuries, fire risk
Electrical Bench	Electric shock, short circuits, tripping hazards	Exposed wiring, overloading circuits	Equipment failure, injuries, possible fire
Software Dev Area	Eye strain, ergonomic injuries, fatigue	Poor posture, long screen exposure	Musculoskeletal disorders, reduced efficiency
Chemical Storage Area	Toxic spill, flammable gas release	Improper labeling, no ventilation	Health risks, explosion, environmental damage
Server Rack	EMF exposure, overheating, cable tripping	Poor thermal management, messy wiring	System failure, personal injury
E-Waste Disposal Zone	Pollution, injuries from sharp objects	Inadequate disposal protocol	Toxic exposure, environmental degradation

CHAPTER FOUR

4. MITIGATION/ELIMINATION STRATEGY(ISO 45001 AND ISO 14001 COMPLIANT)

4.1 MITIGATION/ELIMINATION STRATEGY(ISO 45001 AND ISO 14001 COMPLIANT)

Table 4.2:TABLE OF MITIGATION/ELIMINATION STRATEGY (ISO 45001 AND ISO 14001 COMPLIANT)

Hazard	Mitigation Strategy
Soldering Fumes	Install fume extractors, ensure lab PPE (gloves, goggles), and provide safety training.
Electrical Hazards	Use RCDs (Residual Current Devices), insulate all wiring, and conduct monthly inspections.
Ergonomic Issues	Install ergonomic chairs and desks, promote 60-minute screen breaks, and use monitor risers.
Chemical Exposure	Store chemicals in ventilated, labeled cabinets with MSDS sheets; install fume hoods.
Server Room Risks	Implement cable management systems, temperature sensors, and lockable access control.
E-Waste Handling	Establish labeled bins, partner with e-waste recyclers, conduct regular waste audits.

CHAPTER FIVE

5. ISO 45001 COMPLIANCE CHECKLISTS

5.1 ISO 45001 COMPLIANCE CHECKLISTS

Table 5.3: TABLE OF ISO 45001 COMPLIANCE CHECKLISTS OF AUDIT REPORT OF RISETECH

Number	Requirement	Status(Yes/No)	Evidence Needed
1	Leadership and worker participations	Yes	Safety committees and staff training exist
2	Hazard identification and risk assessment	Yes	Lab-specific risk register maintained
3	Emergency preparedness and response	No	Fire drill record missing
4	Incident investigation and corrective action	Yes	Basic incident logbook in use
5	Use of PPE and controls	Yes	Gloves, goggles, lab coats provided
6	Internal audit and compliance monitoring	No	Periodic audit plan does not form formalized
7	Documentation of procedures	Yes	SOPs displayed in labs
8	Training and awareness programs	Yes	Annual orientation conducted
9	Review and continual improvement	No	No formal review mechanisms in place

CHAPTER SIX

6. ISO 14001 COMPLIANCE CHECKLIST

6.1 ISO 14001 COMPLIANCE CHECKLIST

Table 6.4:TABLE OF ISO 14001 COMPLIANCE CHECKLIST

Number	Requirement	Status(Yes/No)	Evidence Needed
1	Environmental policy documentation	Yes	Displayed near lab entrance
2	Environmental aspect and impact identification	Yes	Waste categorization conducted
3	Legal and compliance obligations	No	No record of updated environmental regulations
4	Operational controls for waste and emissions	Yes	Separate bins for e-waste and chemicals
5	Emergency environmental response	No	No spill containment training document
6	Monitoring and measurement of impact	No	No environmental KPIs defined
7	Waste disposal and segregation procedures	Yes	In practice, but not documented formally
8	Environmental performance review	No	No baseline or improvement benchmarks
9	Employee awareness and training	Yes	General awareness, limited documentation

CHAPTER SEVEN

7. ISO 45001:HAZARDS AND SAFETY REQUIREMENTS

7.1 ISO 45001:HAZARDS AND SAFETY REQUIREMENTS

ISO 45001:2018 is an internationally accepted guideline for effectively managing workplace health and safety risks.

Key Hazards addressed:

- **Physical Hazards:**
The soldering process can result in electric shock, noise, and heat.
- **Chemical Hazards:**
Being exposed to fumes and cleaning agents poses a significant risk to one's health.
- **Biological Hazards:**
Poor equipment hygiene or insufficient sanitation practices can result in health problems.
- **Ergonomic Hazards:**
Extended periods of sitting and repetitive strain injuries are prevalent problems.
- **Psychosocial Hazards:**
Work pressure, stress, and a lack of reporting culture are common problems.



Figure 7.1:HAZARDS ISO45001 HAZARDS OF RISETECH

Safety Requirements under ISO 45001:

- The task entails the adoption of a structured method to identify and evaluate potential risks.
- Wearing personal protective equipment (PPE) is compulsory based on the job requirements.
- Create action plans for handling incidents and evacuating people.
- Promote employee involvement in safety assessments and evaluations.
- Training programs are being put in place to enhance hazard awareness.
- A method for reporting close calls or hazardous situations through communication.

CHAPTER EIGHT

8. ISO 14001:HAZARDS AND ENVIRONMENTAL SAFETY REQUIREMENTS

8.1 ISO 14001:HAZARDS AND ENVIRONMENTAL SAFETY REQUIREMENTS

ISO 14001:2015 is a set of guidelines designed to reduce environmental damage and encourage sustainable practices within organizations.

Key Environmental Hazards Identified at RISETECH:

- **Electronic Waste**
Electronic devices such as old computers, sensors, and batteries have been discovered to contain harmful substances.
- **Chemical Spills:**
The process entails the application of etching chemicals and cleaning agents.
- **Air pollution:**
Toxic Fumes Released During Soldering.
- **Water Contamination:**
Risk of liquid or gas leakage.
- **Energy consumption**
Continuous server operations



Figure 8.2:HAZARDS OF ISO 14001 OF RISETECH

Environmental Safety Requirements under ISO 14001:

- Create a plan for separating and recycling waste.
- The plan entails the installation of spill containment kits and the provision of chemical handling training.
- It entails ensuring proper ventilation in soldering and pcb areas.
- Apply energy-efficient measures for lab equipment that is not in use.
- It entails tracking and evaluating environmental performance indicators, including the amount of electronic waste generated.
- The company is dedicated to ongoing progress by establishing yearly environmental objectives.

CHAPTER NINE

9. CONCLUSION

9.1 CONCLUSION

In conclusion, the HSE audit of RISETECH highlights both the strengths and weaknesses in the existing environmental and safety practices. While there is a decent level of compliance with iso 45001 and iso 14001, there is still room for improvement in terms of documentation, conducting regular internal audits, and formalizing procedures. The organization's safety culture and environmental integrity will be significantly improved through the implementation of performance reviews, formalized training, and waste documentation. By adhering to these global guidelines, RISETECH can create a secure and eco-friendly workplace that fosters creativity and scientific exploration.

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